

Problem

In the periodic function of Fig. 17.62,

(a) find the trigonometric Fourier series coefficients a_2 and b_2 ,

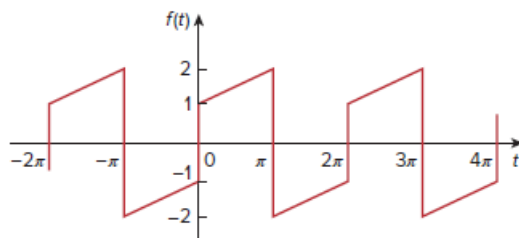
(b) calculate the magnitude and phase of the component of $f(t)$ that has $\omega_n = 10$ rad/s,

(c) use the first four nonzero terms to estimate $f(\pi/2)$,

(d) show that

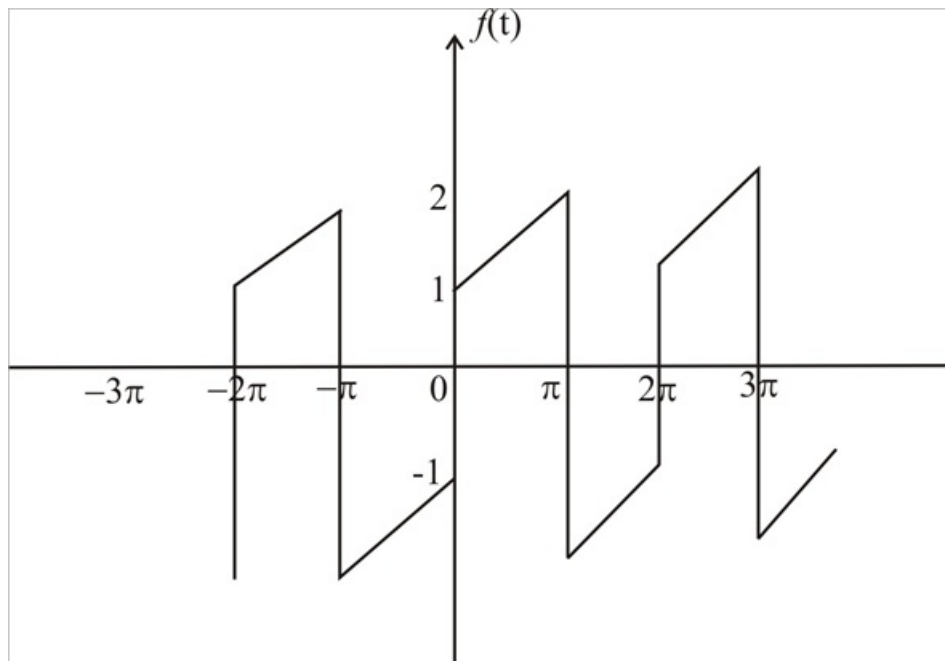
$$\frac{\pi}{4} = \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots$$

Figure 17.62:



Step-by-step solution

Step 1 of 6



Step 2 of 6

Step 3 of 6

Step 4 of 6 (B)
Step 5 of 6 (C)
Step 6 of 6 (D) To show